

Jen-Li Kao

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Perception Systems | Computer Vision & Machine Learning | Signal Processing

EDUCATION

TEXAS A&M UNIVERSITY (TAMU)

College Station, TX

MS in Electrical and Computer Engineering

Sept 2023 – May 2025

- Awards: Graduate Merit Scholarship in Dept of ECE.
- Relevant Coursework: Digital Image Processing & Computer Vision, Pattern Recognition, Data Mining and Analysis, Analysis of Algorithms, MR Engineering, Advanced US Imaging Techniques, Bioelectromagnetism, Algorithm in Structural Bioinformatics

CHANG GUNG UNIVERSITY (CGU)

Taoyuan, Taiwan

MS in Computer Science and Information

Jul 2022

- Thesis: Breast Tumor Detection Using Multi-Channel Millimeter-wave 3D Imaging Technology: A Preliminary Study

NATIONAL DONG HWA UNIVERSITY (NDHU)

Hualien, Taiwan

BS in Computer Science and Information

Jun 2018

PUBLICATION

Conference Paper:

J. -L. Kao and Y. -P. Chao, "*Breast Tumor Detection Using Multi-Channel 62-69 GHz Millimeter-wave 3D Imaging Technology*," 2022 IEEE 4th Eurasia Conference on Biomedical Engineering, Healthcare and Sustainability (ECBIOS), 2022, pp. 16-19, DOI: [10.1109/ECBIOS54627.2022.9945009](https://doi.org/10.1109/ECBIOS54627.2022.9945009)

CONFERENCE

The 16th Symposium of Medical Imaging and Radiological Sciences (SMIRS)., Taoyuan, Taiwan, 22 Oct 2022

- Presentation, "*Breast Tumor Detection Using Multi-Channel Millimeter-wave 3D Imaging Technology: A Preliminary Study*." - Honorable Mention Award

RESEARCH EXPERIENCE

APPLIED MACHINE LEARNING RESEARCH, TAMU

College Station, TX

Research Co-op / Soumyajyoti Dutta (PhD Candidate, TAMU CSE)

May 2025 – Present

Project: Automatic Sigma Rule Generation Using Language and Vision Models

- Developed an end-to-end framework for automated Sigma rule generation by integrating **LLM reasoning**, **AST-based structural parsing**, and **multimodal security inputs**.
- Designed a **language-to-AST parsing pipeline** to evaluate rule consistency, structural complexity (Halstead and Cyclomatic), and syntactic correctness across varied model architectures.
- Prototyping a system that interprets heterogeneous security artifacts—such as log excerpts and visual indicators—and converts them into **machine-interpretable Sigma rules**.
- Advancing to model training and performance evaluation, benchmarking natural-language and AST-guided models on the constructed dataset.

MEDICAL IMAGING PROCESSING AND COMPUTER VISION LAB, CGU

Taoyuan, Taiwan

Research Assistant /Advisor: Prof. Yi-Ping Chao

Sept 2020 – May 2022

Project: Breast Tumor Detection Using Multi-Channel Millimeter-wave 3D Imaging Technology

- Developed a breast tumor detection system using **millimeter-wave imaging** and **computer vision techniques** to **enhance high-resolution, non-invasive cancer imaging**.
- Designed **synthetic 3D phantoms** for breast, tumor, and skin-layer phantoms based on referenced studies to simulate real-world imaging conditions.
- Conducted **6 experiments** to evaluate system performance, including detection depth, minimum tumor size, and multi-angle imaging accuracy to **enhance 3D reconstruction quality**.
- Implemented **noise-reduction** and **beamforming methods**, including averaging and Delay-and-Sum (DAS), to reconstruct enhanced 2D/3D images and **improve overall spatial resolution**.

WORK EXPERIENCE

WE SHARE BASEBALL CO., LTD (WEICHUAN DRAGONS)

Taipei, Taiwan

AI Engineer

Mar 2026 – Present

Project: Vision-Based Player Motion & Injury-Risk Analysis

- Developed computer vision and deep learning pipelines for video-based analysis of pitching mechanics, including pose estimation, temporal feature extraction, and model training.
- Engineered biomechanical features, including joint angles, shoulder external rotation, and limb loading, to quantitatively characterize pitching mechanics.
- Leveraged temporal pose sequences for fine-grained motion classification, supporting injury-risk assessment and return-to-play analysis.

MEDICAL IMAGING PROCESSING AND COMPUTER VISION LAB, CGU

Taoyuan, Taiwan

Research Assistant /Advisor: Prof. Yi-Ping Chao

Jan 2021 – Dec 2021

Project: Mobile Health Technology Medical Device Pre-Market Review Trend Analysis

- Collected and analyzed pre-market guidelines for mobile medical devices and digital health regulations.
- Assisted **ASUS** and **Quanta** with pre-market compliance documents and compiled review Q&A.

CYLTEK LTD.,

Hsinchu, Taiwan

Intern

Sept 2020 – Feb 2021

Project: Smart Healthcare Application Research Project

- Developed Python-based real-time signal-processing algorithms using millimeter-wave radar for contactless measurement of chest displacement, heart rate, and respiration.
- Ported DSP algorithms to C++ for embedded integration and validated data-acquisition timing and synchronization with hardware engineers, alongside MediaTek and Chang Gung Hospital.

TEACHING EXPERIENCE

COURSE: DATA STRUCTURE AND ALGORITHM, DEPT CSIE, CGU

Taoyuan, Taiwan

Teaching Assistant /Instructor: Prof. Yi-Ping Chao

Fall 2021 – Spring 2022

- Assisted in preparing lecture materials, grading, and counseling students on programming issues.

PROJECTS

Benchmarking Transformer-Based Metagenomic Functional Profiling

Spring 2025

- Compared **alignment-based** vs. **ML methods** for enzyme function prediction.
- Accelerated transformer inference with **CUDA** and benchmarked GPU performance on **HPRC** clusters.

Breast Cancer Imaging Classification

Fall 2024

- Conducted a comparative study on **breast cancer classification** using the **mammogram dataset**.
- Built **SVM models with GLCM feature**, compared to **CNN and ResNet-based Transformers**.
- Applied **image preprocessing (normalization, contrast, augmentation)** to improve accuracy.

Integrated MRI Hardware and Pulse-Sequence System Using Analog Discovery 2

Fall 2024

- Built a mini-MRI system using AD2 with **RF control, gradient encoding, and Python acquisition**.
- Calibrated RF and gradient hardware, performed shimming, and acquired optimized spin echoes.
- Developed 2D imaging using **PR** with **gradient encoding, k-space sampling, and FFT**.

TECHNICAL SKILLS

Core Competencies:

- **Imaging & Sensing Systems:** pulse/sequencing design, RF control, gradient encoding, radar sensing
- **Computer Vision & Reconstruction:** multi-dimensional reconstruction, transform-based methods, projection techniques, and data preprocessing
- **Embedded DSP:** adaptive filtering, beamforming, signal synchronization, real-time DSP
- **Machine Learning:** CNNs, Transformers, model optimization, dataset curation, evaluation metrics
- **HPC:** CUDA acceleration, GPU profiling, parallelization, memory optimization

Technical Stack: Python (NumPy, SciPy, OpenCV, scikit-learn, PyTorch, HuggingFace), C/C++, MATLAB, CUDA, TensorFlow, PyWaveforms/ AD2 SDK, Git, Linux, SLURM/HPRC